

Q1)

A)

I, II:

1. Gradient in Y direction, d
2. Shift Right , e
3. Box Blur, b
4. Impulse, f
5. Gaussian smoothing, h

B)

I.

	Canny	LoG
Order of Derivative	1 st	2 nd
Number of Convolution operations	2	1

II.

- (a) , Since the weak edges were detected by the linking step in the Canny Edge Detector

C)

$$R = \det(M) - \alpha \text{trace}(M)^2 = ad - bc - \alpha(a + d)^2$$

$$R = (403 \times 381 - 385 \times 385) - 0.04(403 + 381)^2 = -19268.24$$

R is a Large Negative Number $\rightarrow$ Indicates an Edge

D)

- I. The image at different scales and smoothing used for scale-invariant feature extraction
- II. An approximation of the laplacian used for edge detection with less computational power
- III. They represent a histogram of orientation with each bin spanning 45 degrees